



Observational Analysis of Multithermal Counterstreaming Flows in a Forming Filament and Their Relationship with Local Heating at Filament Footpoints

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Abstract

Utilizing high-resolution imaging and spectroscopic observations from the New Vacuum Solar Telescope, the Interface Region Imaging Spectrograph, and the Solar Dynamics Observatory, we investigated the nature and origin of counterstreaming flows within a forming active region filament. The ever-present counterstreaming flows observed within the filament are identified as interleaved unidirectional mass flows in opposing directions occurring in neighboring threads. Multiwavelength observations corroborate the multithermal nature of these counterstreaming flows: the cool $H\alpha$ component flows at about $10\text{--}20\text{ km s}^{-1}$, while the warm ultraviolet and extreme-ultraviolet components reach $40\text{--}70\text{ km s}^{-1}$. The Si IV 1400 Å line reveals significant microturbulence in the filament's counterstreaming flows, with a nonthermal velocity width of $\sim 40\text{ km s}^{-1}$. These multithermal flows emanate from compact brightenings at the filament's ends, manifesting as small-scale, collimated upflows at their nascent phase. They continuously inject both chromospheric and transition region plasma into the filament channel, thereby feeding the counterstreaming flows. At their base, the Si IV and C II spectral lines show pronounced line broadening and intensity enhancements, indicating significant localized chromospheric heating. Additionally, numerous small-scale photospheric flux emergence and cancellation events, with a magnitude of $\sim 10^{17}\text{ Mx}$, are detected near their base. We suggest that such weak magnetic-field activities, possibly associated with unresolved magnetic reconnection events, drive these persistent upflows and localized footpoint heating. This work elucidates the multithermal origin of counterstreaming flows within a forming filament and provides evidence of localized chromospheric heating at the filament footpoints.

Unified Astronomy Thesaurus concepts: Solar activity (1475); Solar active region filaments (1977); Solar active regions (1974); Solar physics (1476); Solar filaments (1495); Solar magnetic reconnection (1504); Jets (870)

Materials only available in the online version of record: animations

1. Introduction

Solar filaments, also known as prominences when appearing beyond the solar limb, are elongated dark filamentary plasma structures supported by special magnetic field topologies against the solar disk (E. Tandberg-Hanssen 1995). They are always observed above the polarity inversion lines (PILs) of the magnetic field at the photosphere. Historically, they are classified by their evolutionary state of their associated magnetic fields into active-region (AR), quiescent, and intermediate types (S. Parenti 2014). Most filaments observed in $H\alpha$, it is possible to identify a narrow filament main body, known as the spine, running horizontally above the PIL. Long-lived quiescent filaments usually exhibit lateral extensions diverging from the filament spine, known as barbs, though these are not a ubiquitous feature for short-lived AR filaments (see the review of J.-C. Vial & O. Engvold 2015). Earlier views proposed that barbs terminate at the chromosphere, and possibly even reach the photosphere, representing the magnetic-field footpoints of the filaments intersecting the solar surface (i.e., S. F. Martin 1998). However, high-resolution observations suggested that barbs bend down to enter and exit the chromosphere, just close to the photosphere (e.g., J. Chae

et al. 2005; Y. Lin et al. 2005a, 2005b; L. Li & J. Zhang 2013; X.-L. Yan et al. 2015). Typical AR filaments have footpoints only at their extreme ends, whereas longer filaments in ARs or the quiet Sun may possess additional, intermediate footpoints (J.-C. Vial & O. Engvold 2015).

It is widely accepted that filament material originates from the chromosphere, given the similarity in elemental composition and temperature between filament material and the chromosphere (D. Spicer et al. 1998; H. Song et al. 2017). To date, several possible mechanisms are attributed to the origin of filament mass: direct chromospheric mass injection (e.g., Y. M. Wang 1999; J. Chae et al. 2000; J. Wang et al. 2018, 2019), the evaporation–condensation process (S. Antiochos & J. Klimchuk 1991; T. E. Berger et al. 2012; W. Liu et al. 2012; C. Xia et al. 2012; N. M. Viall et al. 2020; B. Yang et al. 2021; H. Chen et al. 2022), and the direct levitation of dipped field lines from the lower chromosphere (D. Rust & A. Kumar 1994; B. Lites 2005; T. J. Okamoto et al. 2008), as well as the lifting mechanism of reconnected magnetic fields (e.g., Y. E. Litvinenko & S. F. Martin 1999; Y. E. Litvinenko & M. S. Wheatland 2005). Recent numerical simulations indicate that as the local heating site ascends, the direct chromospheric mass injection can convert to an evaporation–condensation process (C. Huang et al. 2021).

High-resolution $H\alpha$ observations have revealed that all types of filaments consist of numerous thin and dark thread-like plasma structures when seen against the solar disk.



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Filaments, as a whole, persist as stable structures for hours to days, but individual threads are short-lived (lifetime $\sim$ minutes) and highly dynamic. Each of these thin threads exhibits pervasive and continuous mass flows. B. Schmieder et al. (1991) reported the presence of both blueshifted and redshifted motions within these threads, caused by bidirectional/antiparallel flows. J. Zirker et al. (1998) reported bidirectional/antiparallel flows of $5\text{--}20\text{ km s}^{-1}$ throughout the entire filament structure in both the red and blue wings of the $H\alpha$ line, which they first termed filament counterstreaming flows. Since then, the ubiquitous existence of similar counterstreaming flows in different filaments has been extensively confirmed (Y. Lin et al. 2003, 2005a; J. Chae 2007; O. Panasenco & S. Martin 2008; B. Schmieder et al. 2008; K. Ahn et al. 2010; A. Diercke et al. 2018). Using ultrahigh-resolution observations from the High-resolution Coronal Imager, C. E. Alexander et al. (2013) first reported the existence of extreme-ultraviolet (EUV) counterstreaming flows along AR filament threads within the million-degree corona, with velocities of $70\text{--}80\text{ km s}^{-1}$. The measured velocity of EUV counterstreaming flows is approximately 3 to 4 times higher than that of $H\alpha$ counterstreaming flows.

Counterstreaming flows are now known to be commonplace within all filaments, but the nature and driving mechanism of these flows remain a subject of debate (see the review of P.-F. Chen et al. 2020). In the past, counterstreaming flows have been interpreted as longitudinal oscillations within filament threads, where plasma moves back and forth in gravitational potential wells along the same magnetic field line with differing phases (Y. Lin et al. 2005a). Instead, other researchers believed that counterstreaming flows correspond to unidirectional flows in adjacent threads, where plasma moves along distinct magnetic field lines in opposite directions (X.-L. Yan et al. 2015; P. Zou et al. 2016; H. Wang et al. 2018). Numerous mechanisms have been proposed to explain the origin of such unidirectional flows, including siphonic flows driven by the gas pressure imbalance along flux tubes (P. Chen et al. 2014), opposite oscillation mass flows along distinct field lines (Y. Lin et al. 2003), and small-scale energy releasing events such as weak EUV/UV brightenings (T. Li & J. Zhang 2016; X. Yang et al. 2019b) or recurrent network jets (N. K. Panesar et al. 2020) at magnetic field footpoints. In fact, counterstreaming flows in different parts of a single prominence/filament often demonstrate quite complex mass motion patterns. For instance, Y. Shen et al. (2015) reported that counterstreaming flows in the horizontal prominence foot appear to be driven by the imbalanced gas pressure at both ends, while counterstreaming flows in the vertical prominence foot resulted from the coexistence of continuous upward and downward mass flows due to the collapse of prominence bubbles. X. Yan et al. (2025) recently found that in the upper section of a limb quiescent solar prominence, large-amplitude and long-period longitudinal mass oscillations caused counterstreaming flows, while smaller-amplitude random oscillations and mass injection dominated its middle and lower sections. P. Chen et al. (2014) suggested that counterstreaming flows may result from the combined effect of one or even several of the mechanisms mentioned above, especially the combination of longitudinal oscillations and unidirectional flows in adjacent threads. Indeed, magnetohydrodynamic (MHD) simulation results from Y. Zhou et al. (2020) suggest that if turbulent heating occurs at the footpoints of filament field lines, random

evaporation of material would drive hot unidirectional flows and form cool filament threads; meanwhile, gas pressure driven by turbulent heating can also lead to subsequent oscillations of these threads with random phases, resulting in apparent cool counterstreaming flows.

In this study, we investigate counterstreaming flows in a forming active region (AR) filament using observations from the New Vacuum Solar Telescope (NVST), the Solar Dynamics Observatory (SDO), and the Interface Region Imaging Spectrograph (IRIS). This case study allows us to further investigate the multithermal nature and possible origin of counterstreaming flows in a forming AR filament, utilizing high spatiotemporal resolution observations spanning the chromosphere, transition region, and corona.

The remaining paper is organized as follows: In Section 2, we systematically introduce the observation instruments used in the research institute and the multiband data obtained. In Section 3, based on the processed data, multiband analysis and spectral diagnosis were carried out to reveal the physical connection between the counterstreaming flows and the localized heating at the filament's ends. Section 4 summarizes the key points of this work and discusses their implications.

2. Instruments and Data Analysis

The forming filament under investigation originated within a significantly decayed AR NOAA 12782 located in the southern hemisphere. The Global Oscillation Network Group (GONG; J. Harvey et al. 1996) continuously monitored the formation and evolution of this filament in $H\alpha$. On November 14, this filament was well observed by NVST (Z. Liu et al. 2014) between 01:25 and 05:45 UT and by IRIS (B. De Pontieu et al. 2014) from 02:45 to 03:34 UT on November 14. This temporal overlap in observations provides us with a valuable opportunity to investigate the mass flows and their origin within the forming filament.

We used the NVST $H\alpha$ line-center and the $H\alpha$ off-band images acquired at 6562.8 and $6562.8 \pm 0.6\text{ \AA}$ to study the dynamic characteristics and track the filament footpoints. The spatial resolution of these images is approximately 0.165 pixel^{-1} , with a temporal cadence of 45 s (Z. Liu et al. 2014; X.-L. Yan et al. 2020). To better reveal the mass flow pattern in this forming filament, Doppler proxy maps were constructed using the NVST $H\alpha$ red and blue wing data following the equation: $D = (I_B - I_R)/(I_R + I_B)$, where I_R and I_B represent the absolute pixel intensity observed at the red and blue wings, respectively. All the NVST images are Level 1+ products, in which the dark current and flat field have been corrected, and a speckle masking method was used to calibrate the imaging quality (Y.-y. Xiang et al. 2016). NVST $H\alpha$ line-center and the $H\alpha$ off-band images were first coaligned with each other using the method of Y. Cai et al. (2024), and then registered to a standard calibrated solar full disk via an automatic mapping approach (K. Ji et al. 2019).

IRIS tracked the AR 12782 in a large, dense 320-step raster, resulting in a field of view $167'' \times 175''$. The slit-jaw images (SJIs) observed by IRIS at Si IV 1400, 1330 $\text{\AA}$, and Mg II k 7796 $\text{\AA}$ were used. Their temporal cadence is 37 s and spatial resolution of 0.35 pixel^{-1} . The calibrated Level-2 IRIS spectral data were used, which have already been processed with dark current subtraction, flat-fielding, orbital variation, and geometrical corrections (B. De Pontieu et al. 2014). To characterize the mass flows in the forming filament and

associated local heating at the filament footpoints, the following spectral lines were used in this study: (1) Si IV 1394/1403 Å lines, formed in the transition region (TR) with typical formation temperatures of around 0.08 and 0.15 MK, respectively; (2) Mg II k 2796 Å line, with a formation temperature of 0.01 MK in the chromosphere; and (3) C II 1334/1335 Å lines, with a formation temperature of around 0.25 MK in the upper chromosphere or lower TR. We applied a single Gaussian fit to all the optically thin Si IV 1403 Å line profiles along the slit to determine the line parameters, including peak intensity, line width, and Doppler shift. Spectral lines from nearby neutral and singly ionized atoms in a quiet-Sun region were used for absolute wavelength calibration (e.g., H. Tian et al. 2016; H. Chen et al. 2022).

Additionally, EUV 304 Å imaging data, with a spatial resolution of 0.6 pixel^{-1} and a temporal cadence of 12 s, provided by the Atmospheric Imaging Assembly (AIA; J. R. Lemen et al. 2012), and line-of-sight 45 s magnetograms, with a spatial resolution of 0.5 pixel^{-1} , from the Helioseismic and Magnetic Imager (HMI; P. H. Scherrer et al. 2012) onboard the Solar Dynamics Observatory (SDO; W. D. Pesnell et al. 2012) were also used. To compensate for solar rotation, all the SDO, NVST, and IRIS/SJI images taken at different times were aligned to an appropriate reference time of 01:35 UT.

3. Results

Figure 1 depicts the formation and subsequent growth of this target AR filament. From 2020 November 13 through 2020 November 16, this filament started to form and evolved into a typical AR filament. Note that on November 13, the target presented as a collection of chromospheric dark threads exhibiting characteristics akin to an arch filament system, overlying the PIL at a relatively large angle. By November 15, it had gradually evolved into a typical filament, aligned along the PIL with a discernible increase in its apparent length exceeding 30 Mm. Thus, we termed it a nascent filament in the current study.

With the multipassband H α monitor of NVST on November 14, this forming filament was found to be a highly dynamic structure, consisting of relatively long dark threads (see Figure 1(f)). As illustrated in Figures 1(g) and 2(a)–(b), distinct upflows and downflows coexist along the filament axis. These mass flows well traced the magnetic field morphology of this forming filament, particularly its footpoints. As indicated by the green and yellow arrows in Figure 1(g), the south footpoints (SF) of this forming filament were rooted in the positive-polarity magnetic field region of the AR, and the north footpoints (NF) were located in the negative-polarity magnetic field flux of the AR (also see red ellipses in Figure 1(i)). In accordance with the chirality definition proposed by S. F. Martin (1998), it is categorized as a sinistral filament.

Based on its footpoint regions (SF and NF), it can also be noted that the forming filament appears to be a high-lying structure, with its main axis arching at a relatively large angle (about 30°) to the PIL of the decayed AR. This apparent angle of the filament relative to the PIL is mainly due to a prominent projection effect on November 14. Moreover, the filament might still be in a nascent state, with its threads gradually aligning along the PIL. After careful inspection of the associated movie, no obvious helical or twisted trajectory of

mass flows was discerned within this forming filament, suggesting a possible sheared arcade magnetic configuration. Such sheared arcade magnetic configurations appear to be commonly found in forming AR filaments or reforming intermediate filaments, as suggested by previous studies (e.g., Y. Ouyang et al. 2017; H. Chen et al. 2018).

As shown in Figure 1(g) and the associated animation, persistent blueshift and redshift coexisted along the filament's main axis. At certain times, blueshift and redshift overlapped or appeared at different ends of the filament (see Figure 1(g)), while at other times they appeared along adjacent filament threads (see Figure 2(c)). These signals indicate the occurrence of filament counterstreaming flows, a long-lived phenomenon that persisted throughout the formation and evolution of this filament from 01:25 UT to 05:45 UT. To characterize these H α component upflows and downflows, two time–distance plots were generated along slices AB and CD in Figures 2(a) and (b). The results are presented in Figures 2(d) and (e), respectively. In the H α red wing (6563.6 Å), significant downflows, with mean projected velocities ranging from -23 to -10 km s^{-1} , drained from the NF to the SF. In the H α blue wing (6562.4 Å), upflows with velocities ranging from 9 to 26 km s^{-1} were observed, streaming from SF to NF. The most striking feature observed here is the presence of upward mass flows only in the H α blue wing and downward mass flows only in the H α red wing. This result strongly supports the view that counterstreaming flows in this forming filament represent interleaved unidirectional mass flows in opposing directions along neighboring filament threads, rather than filament mass longitudinal oscillation. As shown in Table 1, the mean velocity of these downflows/upflows detected in Figure 2 is -18 and 16 km s^{-1} , respectively. The velocities of these downflows and upflows fall within the typical velocity range of previously observed H α counterstreaming flows (e.g., J. Zirker et al. 1998; Y. Lin et al. 2005a).

Figure 3 shows the IRIS spectroscopic observation of this forming filament during 02:45–03:34 UT. As illustrated in Figure 3(c), the ongoing counterstreaming flows are also evident in the Doppler shift map of the Si IV 1402.77 Å line. Its global Doppler shift pattern well aligns with that revealed in the NVST H α Doppler proxy map. The absolute Doppler shift velocities of these Si IV counterstreaming flows range from 20 to 45 km s^{-1} , which is significantly higher than the flow velocities observed in H α passbands. Meanwhile, Figure 3(b) reveals a greatly enhanced line width velocity, with a typical value of 40 km s^{-1} , within the forming filament. Given a thermal broadening velocity of 6.8 km s^{-1} and an instrumental broadening velocity of 4.1 km s^{-1} (e.g., H. Tian et al. 2014; J. Zhang et al. 2017), the nonthermal width velocity in the forming filament is of the same order. It may be caused by the significant microturbulence generated by antiparalleled mass flows. Furthermore, using the wavelength scan of the Mg II k 2796.35 Å line, Figures 3(e)–(h) reveal the fine-scale motion structures of the forming filament across different velocity ranges spanning the lower to upper chromosphere. In the red wing of the Mg II k line, evident mass flows drain downward along the forming filament toward its south end (as marked by the white arrow pair in Figure 3(g)), within the redshifted velocity range of -20 to -60 km s^{-1} . In the blue wing of the Mg II k line, sporadic and threadlike upflows emanate from the filament's south end (as marked by the yellow arrows in Figure 3(e)), in the blueshifted velocity range

Table 1
Detailed Information of the Upflows/Downflows Detected at Different Imaging Passbands

Mass Flows	Event No.	Start Time (UT)	Duration (minutes)	Velocity (km s^{-1})	Band ($\text{\AA}$)
Downflows along the filament spine detected in Figure 2(d)	1	01:50	25	−10	NVST $\text{H}\alpha$ +0.6
	2	02:55	10	−18	...
	3	03:10	15	−18	...
	4	03:32	11	−21	...
	5	04:20	6	−20	...
	6	04:50	5	−23	...
	7	05:03	8	−15	...
Upflows along the filament spine detected in Figure 2(e)	1	01:32	8	15	NVST/ $\text{H}\alpha$ −0.6
	2	01:50	6	18	...
	3	01:56	15	9	...
	4	02:26	5	26	...
	5	03:00	10	11	...
	6	03:08	6	23	...
	7	03:14	9	17	...
	8	03:59	4	20	...
	9	04:26	9	10	...
	10	05:05	7	14	...
	11	05:07	16	14	...
	12	05:25	5	18	...
Upflows at the filament's end detected in Figure 4(d)	1	03:09	>3	38	NVST/ $\text{H}\alpha$ −0.6
	2	03:13	>2	20	...
	3	03:14	>2	27	...
	4	03:17	>2	34	...
	5	03:18	>2	26	...
Upflows at the filament's end detected in Figure 4(e)	1	03:04	2	29	SDO/AIA 304
	2	03:10	1	44	...
	3	03:11	2	14	...
	4	03:17	0.5	69	...
	5	03:18	3	27	...
	6	03:25	1	59	...
Upflows at the filament's end detected in Figure 4(f)	1	03:10	1	79	IRIS/SJI 1400
	2	03:23	1	34	...

Note. Duration: the time period spanning from the initial appearance to the final disappearance of the upflows or downflows, determined from the corresponding time–distance plots. The duration of the $\text{H}\alpha$ −0.6 upflows in Figure 4(d) should be considered lower limits, due to the shorter slice path of the time–distance plots. Velocity: upward flows are assigned a positive value, while downward flows are assigned a negative value. Along the filament spine, downflows and upflows in $\text{H}\alpha$ passbands have a mean velocity of −18 and 16 km s^{-1} , respectively. At the filament's south end, upflows have a higher mean velocity: 29 km s^{-1} in $\text{H}\alpha$ passband and 44 km s^{-1} in EUV/EUV passbands.

of 32–70 km s^{-1} . Combined with the NVST observations, this result indicates counterstreaming flows in the forming filament occur in both chromospheric and transition region temperatures.

The peak intensity image of the Si IV 1402.77 $\text{\AA}$ line in Figure 3(a) reveals a series of compact UV brightenings appearing at the SF and NF regions of this forming filament (as denoted by black and white circles). The IRIS slit crossed some of these compact UV brightenings. Figures 3(i)–(k) present the spectral profiles of the Si IV 1402.77 $\text{\AA}$ line, the C II line pair (1334.5 and 1335.7 $\text{\AA}$), and the Si IV 1394 $\text{\AA}$ line extracted from the NF and SF regions of the filament (see red and blue boxes in circles). Compared with the reference spectra extracted from a nearby plage region (indicated by the black dashed-line box in Figures 3(a) and (e)), spectral lines in the SF and NF regions reveal a clear enhancement and evident line broadening in both Si IV and C II lines, while still maintaining a Gaussian profile. Notably, the C II line pair in the NF region shows a reversal at the line core, which is absent in the SF region. Moreover, similar compact footprint

brightenings were also detected in AIA EUV 304 $\text{\AA}$ observations (Figure 1(h) and the associated animation). In Figure 2(e), the normalized 304 $\text{\AA}$ AIA flux, calculated within the SF region (denoted by the box in Figure 1(h)), indicates a series of repeated enhanced flux peaks. These repeated EUV brightenings appear to be associated with the temporal occurrence of episodes of $\text{H}\alpha$ upflows, repeatedly feeding mass supply into the forming filament.

Indeed, a closer look at the SF region of the filament indicates such compact brightenings associated with a series of small-scale upflows, as marked by the yellow arrows in Figure 3(a). These small-scale upflows, known as network jets, were first reported by H. Tian et al. (2014) using IRIS SJI observations with the 1300 and 1400 $\text{\AA}$ filters. They are often preceded by compact network brightenings, representing short-lived, high-speed upflows, with speeds ranging from 80 to 200 km s^{-1} and temperatures of at least $\sim 10^5$ K. Figure 4 presents one selected episode of such small-scale jets at the SF region. The multiwavelength images from NVST, IRIS, and AIA reveal that these small-scale jets exhibit multitemperature

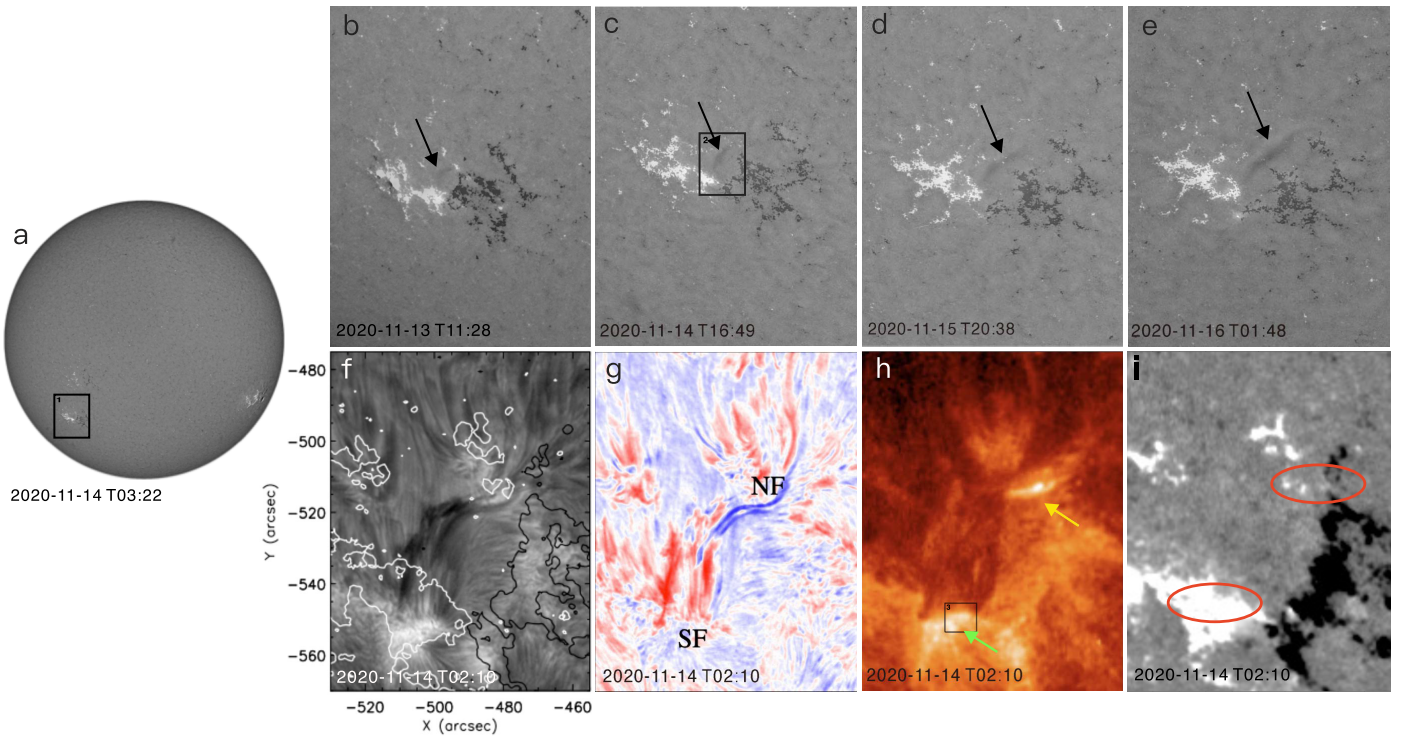


Figure 1. Overview of the target filament. Panels (a) to (e) were composite images of GONG H α and SDO/HMI line-of-sight magnetograms. The black box in panel (a) marks a smaller field of view (FOV) containing the target filament of panels (b) to (e). Panels (b) to (e) show the forming filament observed on November 13, 14, 15, and 16, respectively. Black arrows indicate the forming filament. The black box in panel (c) provides the FOV of panels (f) to (i). Panels (f) and (g) show the NVST H α line center and Doppler proxy map constructed from the H α red and blue wing images. Panels (h) and (i) present the simultaneous AIA 304 Å image and HMI line-of-sight magnetogram. The simultaneous HMI line-of-sight magnetogram, with the level of ± 100 G, was overplotted in panel (f). Yellow and green arrows indicate the footpoint regions of the filament. Red ellipses denote their magnetic environment. An animation of panels (f)–(i), with a duration of 15 s, is available.

(An animation of this figure is available in the [online article](#).)

components (as marked by arrows), which is in line with the very recent observational results of network jets (T. Samanta et al. 2019; Y. Duan et al. 2025; E.-K. Lim et al. 2025). In H α blue wing images, these cool jet components appeared to be taller and darker upflows, resembling the enhanced spicules reported by T. Samanta et al. (2019) and macrospicules reported by Y. Duan et al. (2023), while they are less continuous in AIA 304 Å passbands. The time–distance plots shown in Figures 4(d)–(f) clearly demonstrate that these small-scale jets are driving multithermal mass into the filament’s end along the slice path. Their cooler H α jet components exhibit lower ejection velocities of $20\text{--}38\text{ km s}^{-1}$ (with a mean of 29 km s^{-1} as shown in Table 1). In contrast, the warmer UV/EUV jet components are ejected with higher velocities, ranging from $30\text{--}79\text{ km s}^{-1}$ (with a mean of 44 km s^{-1} in Table 1). Compared to the H α upflows detected along the filament spine (which have a mean velocity of 16 km s^{-1}), the cool jet components newly initiated at the filament’s end exhibit a larger velocity and possibly no deceleration.

Figure 5 presents the EUV brightening and associated small-scale jets at the NF region. As shown in Figures 5(a) and (b), the EUV brightening at the NF region occurred below the north end of the forming filament, located at a magnetic mixed-polarity region. At this location, small-scale opposite-polarity magnetic fluxes converged and canceled with each other, resulting in an obvious decrease of negative flux in its source region (see Figure 5(i)). Using the NVST red wing observations, we identified a minifilament, with a projected length of approximately 10 Mm, at the same location prior to the EUV

brightening at 01:35 UT. The minifilament began to erupt at approximately 01:40 UT. Its eruption then led to the EUV brightening in the AIA 304 Å passband and an obvious rolling motion along the long thread of the forming filament in the NVST red wing images (as indicated by the blue circular arrow in Figure 5(d)). Subsequently, a small-scale jet occurred following this rolling motion, injecting mass into the forming filament. Such apparent rolling motions are most likely triggered by a rapid twist transfer from the eruption minifilament into the forming filament. Similar phenomena have been reported in the filament eruptions in the footpoints of coronal loops (e.g., J. Liu et al. 2019; L. Yang et al. 2019a; H. Chen et al. 2021; X. Zhou et al. 2024). To characterize the mass motion associated with this jet, two time–distance plots were generated along slices GH and JK, respectively. As shown in Figure 5(g), this small-scale jet manifested as bright EUV upflows, with measured velocities of $19\text{--}34\text{ km s}^{-1}$, in the AIA 304 Å passband. Subsequently, antiparallel dark EUV downflows drained out along the same path, with velocities of $39\text{--}59\text{ km s}^{-1}$. In the H α passband, the jet appeared as cool mass upflows and downflows, with velocities ranging from 11 to 14 km s^{-1} . It is important to note that the H α upflows/downflows were, respectively, detected in the red/blue wing passbands due to the projection effect in the southern hemisphere.

Flux cancellation refers the mutual close and then disappearance of opposite-polarity magnetic fragments in the photosphere, which has been suggested to be signatures of magnetic reconnection at lower atmosphere (C. Zwaan 1987, and its references). Extensive observations reveal that flux

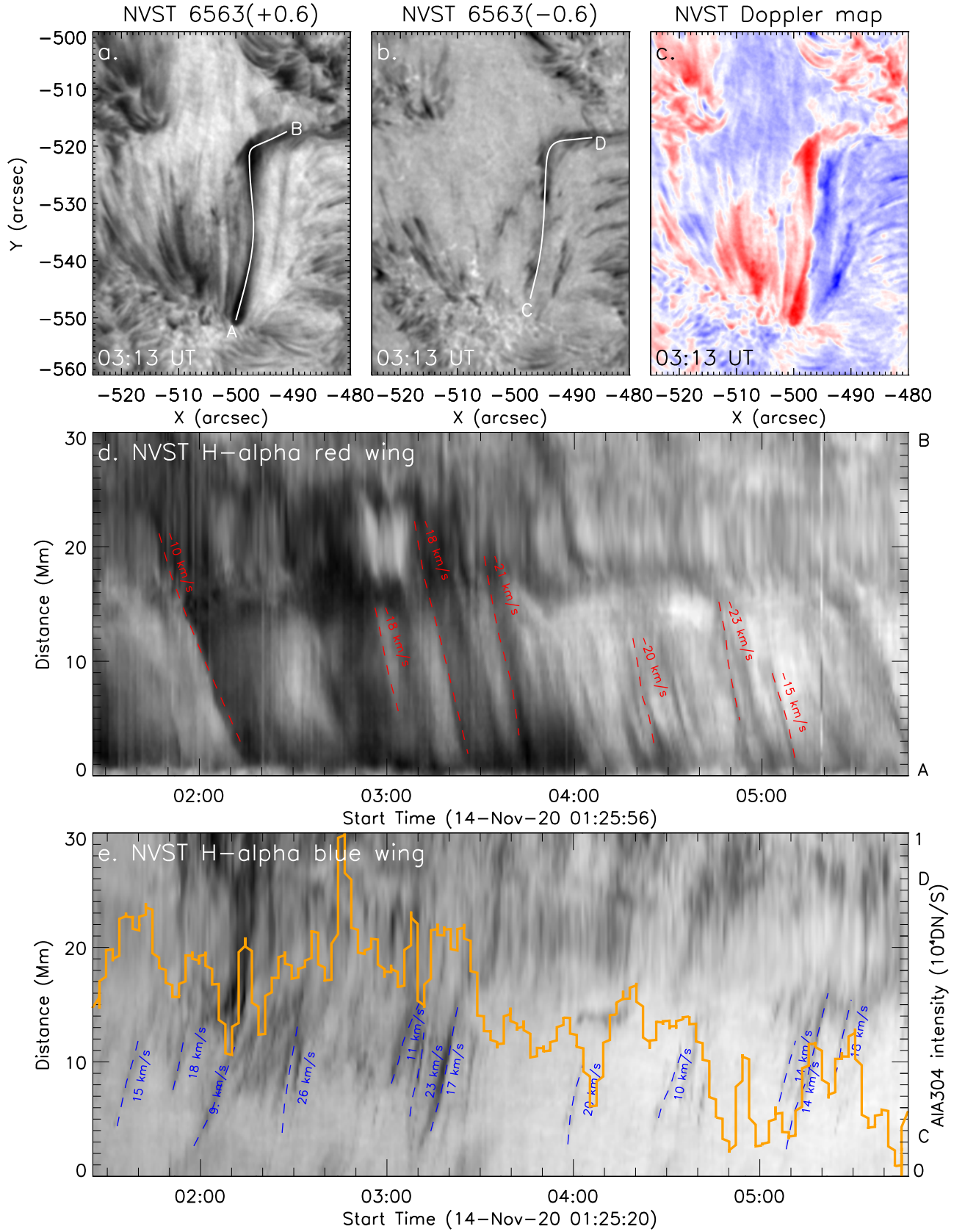


Figure 2. The mass flows in the forming filament revealed by the NVST observations. (a) The red wing of H α ; (b) the blue wing of H α ; and (c) the Doppler proxy map, which combines (a) and (b) in their corresponding colors. Panel (d) shows the time-distance plot along the path AB, indicated in panel (a), with a downward average velocity of approximately 18 km s^{-1} . Panel (e) shows the time-distance plot along the path CD indicated in panel (b), with an upward average velocity of approximately 16 km s^{-1} . Meanwhile, the normalized $304 \text{ \AA}$ EUV intensity flux at the south footpoint is overplotted as an orange curve in panel (e). The animation of this figure demonstrates the counterstreaming flows observed by the NVST. The animation's duration is 15 s.

(An animation of this figure is available in the [online article](#).)

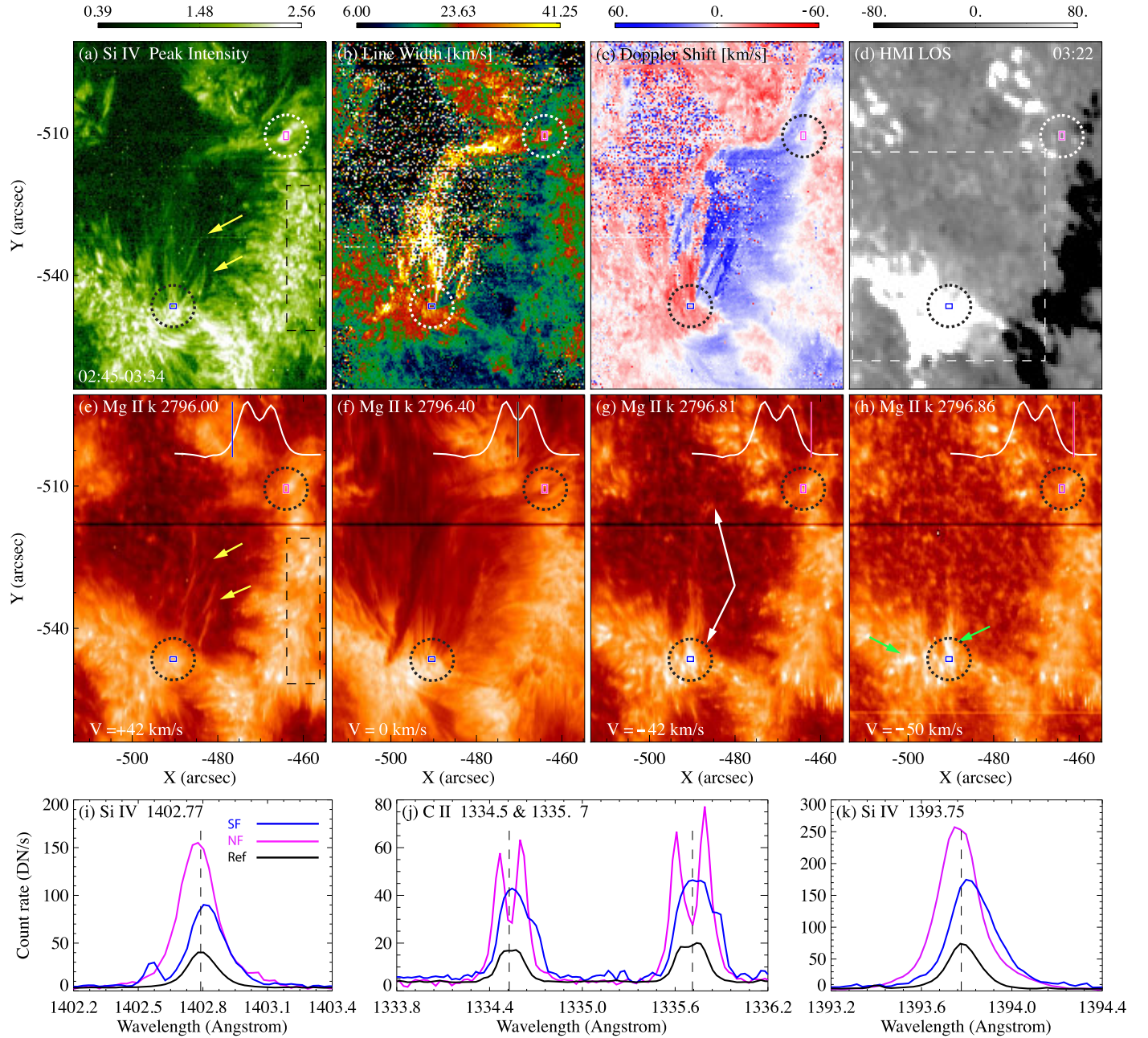


Figure 3. Interface Region Imaging Spectrograph (IRIS) spectroscopic observations of the forming filament during 02:45 to 03:34 UT. (a)–(c): The peak intensity, line width, and Doppler shift of the Si IV line at 1393.75 Å, obtained by a single Gaussian fit. (d) The HMI LOS magnetogram, with the level of ± 80 G, at 03:22 UT. The white dashed-line box denotes the FOV of Figure 6(a). (e)–(h): Snapshots of Mg II k line at different wavelength from blue wing to red wing, showing the mass flows along the forming filament at different velocities. Blue boxes in Figures (a) to (h) mark the areas where the spectral lines at the southern footpoint of the filament are extracted and spatially averaged, and the purple boxes mark the areas at the northern footpoint. It is marked with a more obvious black and white dotted circle in the picture. The black dashed boxes in (a) and (e) are the selected reference patches. (i)–(k): Spectral lines of Si IV 1402.77 Å, C II 1334.5 and 1335.7 Å, and Si IV 1393.75 Å at the south footpoint region (blue box), the north footpoint region (purple box) of the forming filament, and the reference plage region (black dashed-line box), respectively. Yellow arrows in panels (a) and (e) indicate numerous small-scale jets, representing threadlike upflows along the forming filament. The white arrow pair in panel (g) indicates mass flows within the forming filament, with a distinct brightening region at its south footpoint. Green arrows in panel (h) indicate brightenings in the low chromosphere. An animation demonstrates the counterstreaming flows and footpoint region brightenings revealed by the IRIS wavelength scan of the Mg II k line. The animation’s duration is 14 s.

(An animation of this figure is available in the [online article](#).)

cancellation are closely related to the formation and eruption of minifilaments (e.g., J. Hong et al. 2011; H. Chen et al. 2020, 2025) and the solar jet activities, i.e., coronal jets (e.g., A. C. Sterling et al. 2016, 2017; N. K. Panesar et al. 2018; J. Yang et al. 2023), network jets/jetlets (e.g., N. K. Panesar et al. 2020; Y. Duan et al. 2025; L. Yang et al. 2025), as well

as enhanced chromospheric spicule activities (T. Samanta et al. 2019; Y. Duan et al. 2023; A. Bura et al. 2025). From Figures 5(g) and (h), it is evident that the minifilament-induced EUV brightening and mass flows both occurred at the north filament’s end during the period 01:45–02:00 UT. Obvious flux cancellation occur below the minifilament site. In

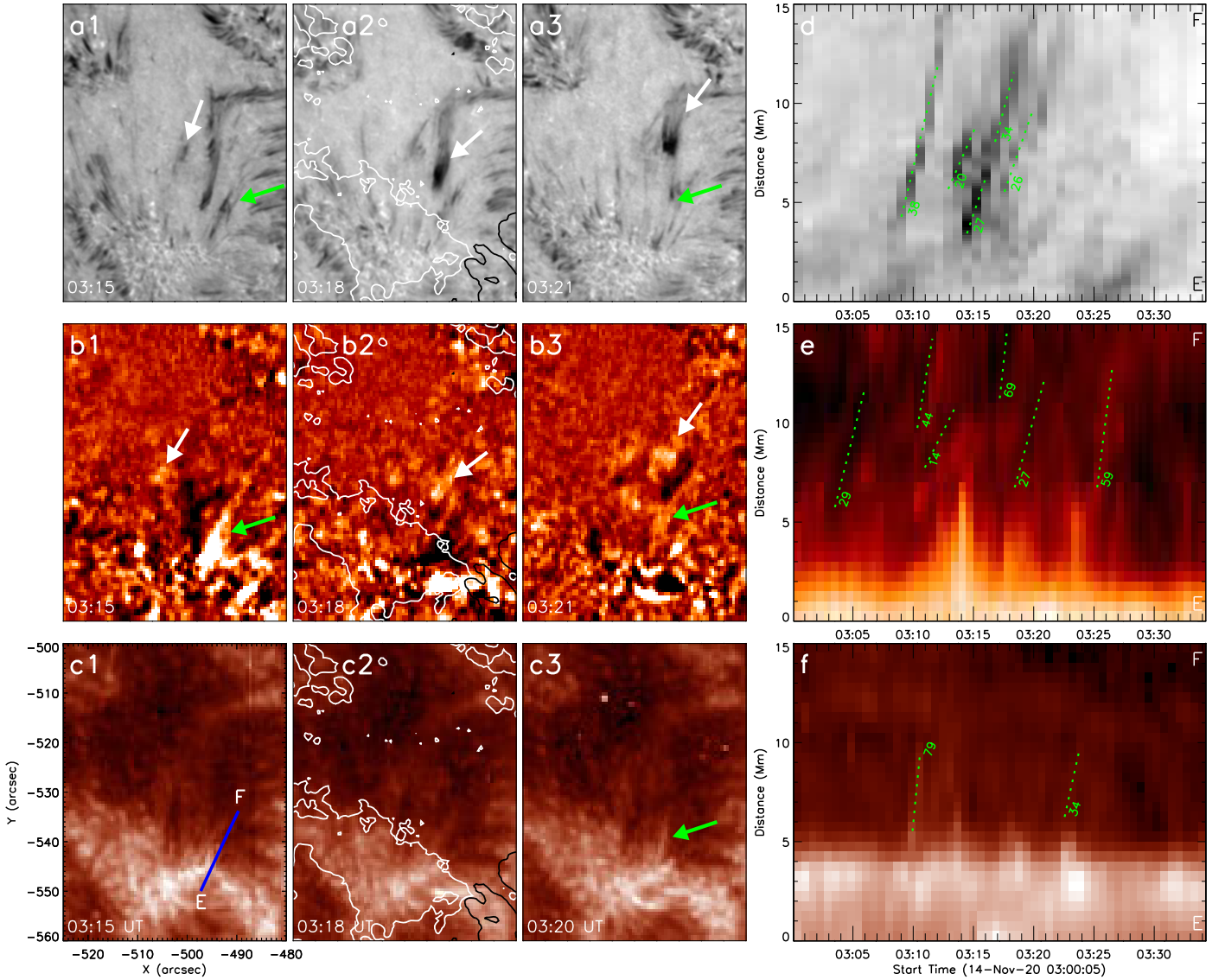


Figure 4. Multitemperature nature of a selected episode of small-scale jets observed at the south footpoint of the forming filament. Panels (a), (b), and (c) give the NVST $H\alpha$ blue wing images, AIA 304 Å images, and IRIS/SJI 1400 Å images, respectively. HMI contours of level ± 30 G are overlaid in the second column. The green and white arrows mark multitemperature components of several ongoing small-scale jets. (e)–(f): Time–distance plots of NVST $H\alpha$ blue wing, AIA 304 Å, and IRIS/SJI 1400 Å images along the slice EF in (c1). An animation of this figure is available, showing the occurrence of small-scale jets. The duration of the animation is 3 s.

(An animation of this figure is available in the [online article](#).)

Figure 5(i), the AIA 304 Å flux and magnetic flux variation were calculated within the source region of the erupted minifilament (as indicated by the blue dashed box in Figures 5(a) and (c)). Before 1:40 UT in the source region of filament, both the positive and negative magnetic fluxes were decreasing. After that, the negative flux decreased significantly, while the positive flux increased slightly. This process indicates that the cancellation of the magnetic flux occurred, with a magnitude of 10^{17} Mx. Similar flux cancellation at smaller scales also occur at the filament's south end. Figures 6(a)–(c) and associated animation present the photospheric magnetic evolution at the SF region. In fact, small-scale jets at SF region originated from a positive-polarity magnetic field region, but in which numerous very weak, small-scale negative-polarity flux elements randomly emerged. Such random negative-polarity flux emergences are only

noticeable when the magnetic level of the HMI LOS magnetograms set as ± 20 G. As marked by the black arrows in Figures 6(b) and (c), their spatial scale ranges from approximately 2 to 5 pixels, namely 0.7 to 1.8 Mm, which approaches the spatial resolution limit of the HMI. At the SF region, these small-scale flux elements rapidly emerged and soon disappeared within several minutes, occurring near the starting points of the small-scale jets and blueshift sources. Figure 6(d) shows the flux variation of negative-polarity elements in three selected box regions (S1, S2, and S3 denoted as blue dashed boxes in Figure 6(a)). The magnetic flux variation is on the order of 10^{17} Mx, indicating a very weak flux emergence and possible subsequent cancellation. Consistent with the views mentioned above, these results suggest that flux cancellation may play an important role in triggering these small-scale jets at the filament's ends.

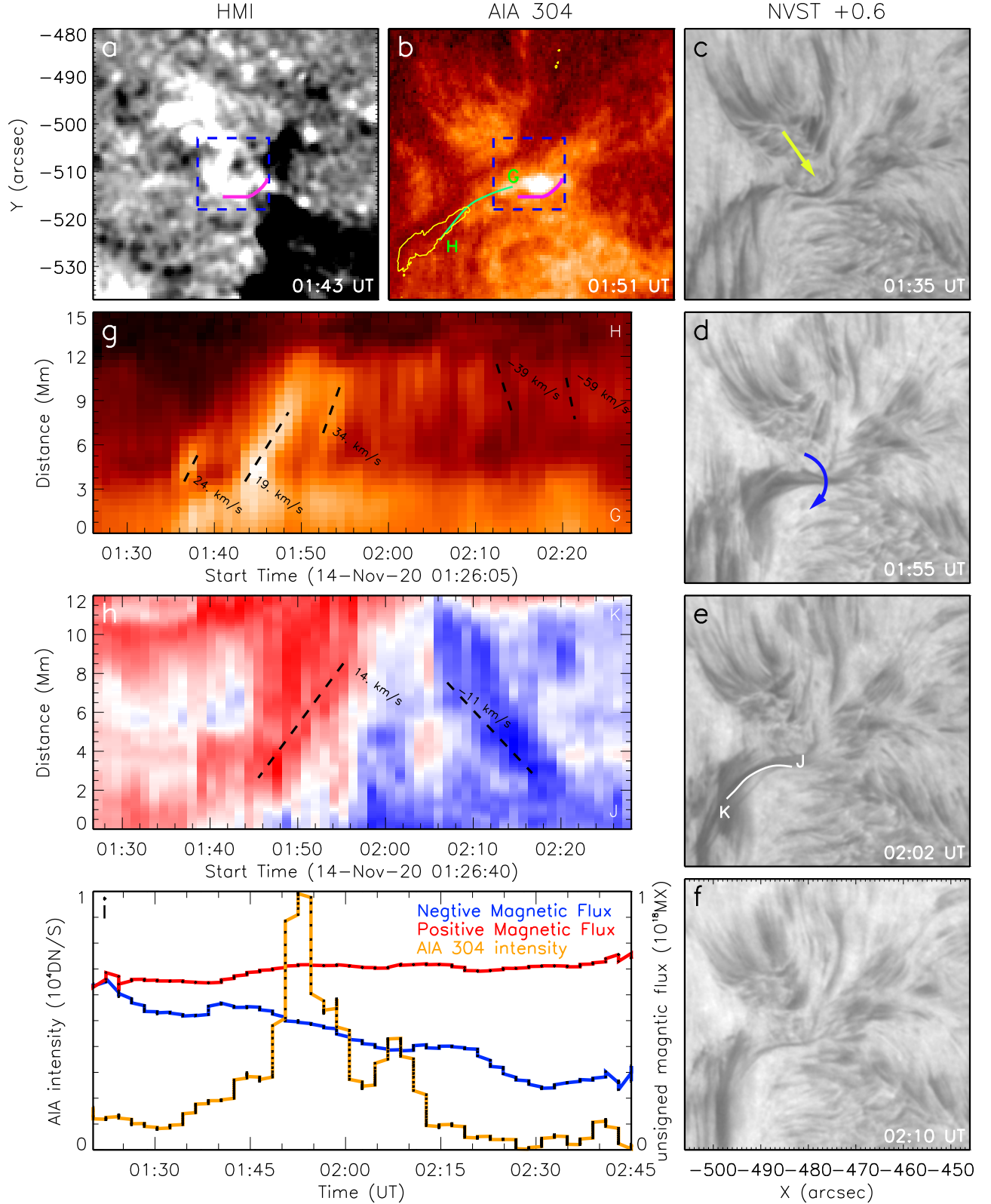


Figure 5. Small-scale jets and associated minifilament eruption observed at the north footpoint of the forming filament. Panels (a) and (b) show the SDO/HMI line-of-sight magnetogram and the AIA 304 Å image near the northern footpoint of the filament. The blue dashed box shows the area selected for the calculation of the magnetic flux. The yellow contour represents the outline of the forming filament observed in the NVST H α line core. Panels (c)–(f) show NVST H α red wing images of the same FOV, demonstrating the minifilament eruption and associated untwisting motion. Panel (g) is the time–distance plot along the path GH in panel (b), showing upward and downward flows. Panel (h) presents the time–distance plot derived from Doppler proxy images along the spatial path JK indicated in panel (e), wherein signals corresponding to both the red and blue wings of the spectral line are displayed. Panel (i) shows the magnetic flux variations and the normalized radiative intensity changes of AIA 304 Å. An animation shows the minifilament eruption and magnetic flux variation in its source region. The duration of the animation is 2 s.

(An animation of this figure is available in the [online article](#).)

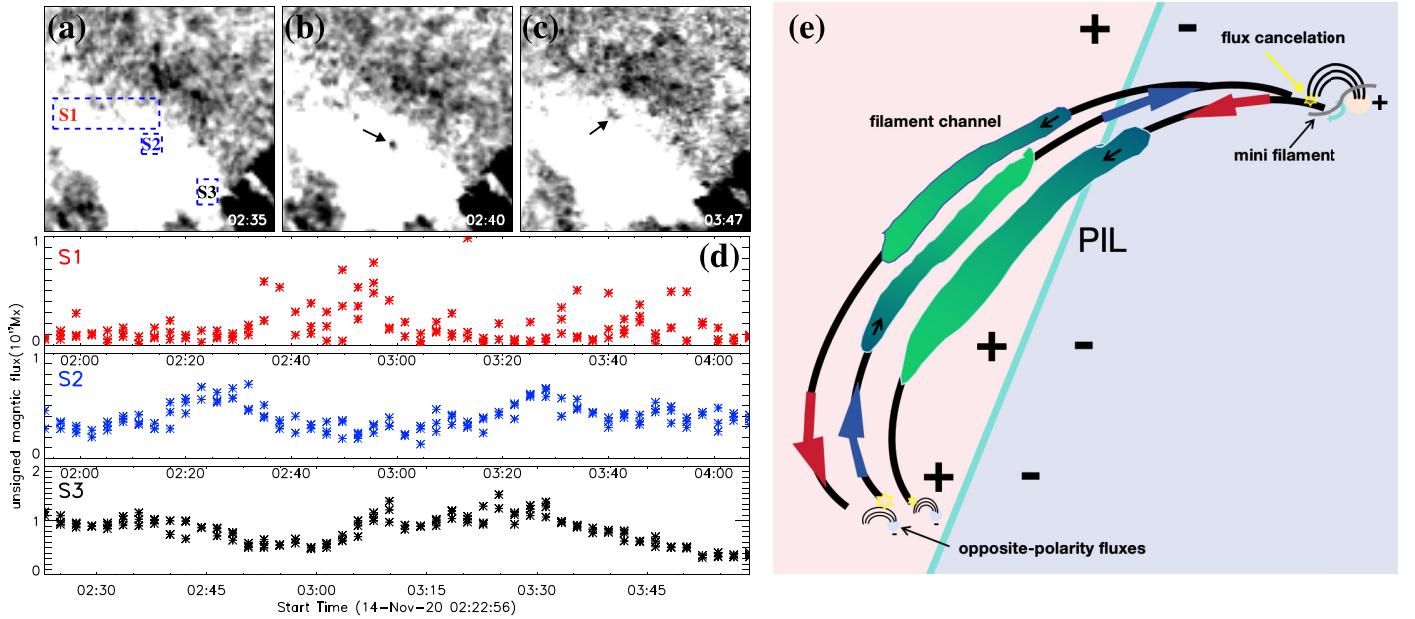


Figure 6. (a)–(c) Snapshots of selected HMI LOS magnetograms with a magnetic level of ± 20 G. In panel (a), three small regions, S1, S2, and S3, near the southern footpoint were selected to calculate their magnetic flux. (d) The magnetic flux evolution in these three regions. (e) A schematic diagram illustrating the proposed scenario, explaining the nature and origin of the forming filament based on observations. An animation is available, showing the occurrence of emergence and disappearance of small-scale negative-polarity magnetic flux elements. The duration of the animation is 5 s.

(An animation of this figure is available in the [online article](#).)

4. Discussion

Counterstreaming flows are a prevalent intrinsic motion within filaments, and understanding their observational characteristics and physical origins is a fundamental problem. Over the past decades, extensive solar imaging observations have revealed the frequent occurrence of jet activity along the legs of coronal loops or at filament footpoints (e.g., J. Chen et al. 2015; A. C. Sterling et al. 2016; J. Hong et al. 2017; J. Wang et al. 2019; L. Yang et al. 2019a). Consequently, some researchers proposed that jet-like mass ejections may be the direct cause of counterstreaming flows in filaments (Y. M. Wang 1999; J. Chae et al. 2000); however, this hypothesis has hitherto lacked robust observational support (see D. H. Mackay et al. 2010). Recently, C. Zhou et al. (2025) reported that recurrent coronal jets at one end of a solar prominence induced both forward- and backward-moving mass flows along the adjacent magnetic field lines of the prominence cavity, consequently giving rise to a Doppler bullseye pattern within the cavity as observed from the limb. Recent observations by N. K. Panesar et al. (2020) provided solid evidence that recurrent network jet eruptions directly drive EUV counterstreaming flows along filament/filament channel threads. Compared with their observations, our current findings further reveal that: (1) counterstreaming flows along the threads of a forming filament/filament channel comprise plasma ranging from chromospheric to transition region temperatures; (2) the cool and warm components of the counterstreaming flows are directly driven by a series of small-scale jets emanating from chromospheric and transition region heights; and (3) UV/EUV brightenings near the SF and NF regions, as revealed by IRIS spectroscopy, indicate local heating at the filament footpoints. Regarding the origin of these small-scale jets, our observations also support the view that flux cancellation results in the occurrence of small-scale jets at both the SF and NF regions. Specifically, for the jets

observed in the NF region, high-resolution chromospheric observations from NVST revealed the presence and subsequent eruption of a small minifilament after flux cancellation. Consistent with previous observations (e.g., N. K. Panesar et al. 2018; H. Chen et al. 2020), this result suggests that flux cancellation may be the primary candidate for triggering the minifilament eruption and its associated jet activity. This eruption not only injected its mass but also transferred its magnetic twist into the developing AR filament. A notable feature of the current study is the occurrence of small-scale jets in the SF region within a predominantly positive-polarity plage region. At the detection limit of the HMI magnetic field resolution, we identified numerous small-scale negative-polarity flux elements emerging and disappearing with a magnitude of $\sim 10^{17}$ Mx. This implies that flux emergence and cancellation occurring near or even below the resolution limit of HMI may be present in the base region of these small-scale jets. Therefore, we conjecture that smaller-scale opposite-polarity flux elements and their associated flux cancellation processes may drive these small-scale jets observed in the SF region.

Based on our results described above, a possible scenario was proposed in Figure 6(e) to interpret the current observations. The forming filament arcade above the PIL, with its south footpoints anchored at a predominantly positive-polarity plage region. The emergence and subsequent disappearance of negative-polarity minor magnetic polarities may trigger repeated magnetic reconnection events (illustrated as a small arch in Figure 6(e)), resulting in a series of small-scale jets and UV/EUV compact brightenings. These recurrent jets inject multitemperature plasma upward into the filament, continuously supplying the necessary material and energy to sustain the observed counterstreaming flows. At the filament’s northern footpoint, magnetic flux converges and cancels in the photosphere. This

process led to the formation and eruption of a tiny minifilament. Its eruption undergoes unwinding, injecting both twist and material into the forming filament. Due to the occurrence of recurrent small-scale jets and associated local heating, continuous mass flows form along different nearby filament threads. Notably, these mass flows, characterized as Doppler red/blueshifts, occur sequentially along individual magnetic field lines and coexist simultaneously along different field lines within the filament, eventually causing the apparent counterstreaming flows.

Local heating at the footpoints of magnetic fields is a crucial input in MHD numerical simulations of solar prominences/filaments and coronal rain (e.g., C. Xia et al. 2012; C. Xia & R. Keppens 2016; V. Jerčić & R. Keppens 2023), but observational constraints on the occurrence and nature of this local heating are currently lacking. Within the chromospheric evaporation–coronal condensation framework, researchers have typically assumed localized random heating at the footpoints of magnetic arcades, distributed randomly both spatially and temporally, with the heating being 1 to 3 orders of magnitude stronger than the background heating (e.g., C. Xia et al. 2011, 2012; Y. Zhou et al. 2020; C. Zhou et al. 2025). Using high-resolution imaging in He I 10830 Å and H α observations of a set of AR loops associated with counterstreaming motions, X. Yang et al. (2019b) reported weak EUV brightenings in intergranular lane areas and suggested that these subtle EUV brightenings are a possible localized heating signal at the footpoints of AR loops. T. Li & J. Zhang (2016) reported the detection of subarcsecond UV bright points located beneath a quiescent prominence, and suggested that these brightening features, in conjunction with associated quasiperiodic upflows, may facilitate the transport of mass into the overlying large-scale prominence. In this work, our observations provide direct IRIS spectral and imaging evidence of local heating at the filament footpoints. At the compact brightenings in the SF and NF regions, spectral profiles of Si IV and C II lines reveal a clear enhancement and evident line broadening. The increase in the intensity of both Si IV and C II lines may be attributable to localized thermal enhancement of cooler plasma within the upper chromosphere. The line broadening of the Si IV and C II lines was possibly caused by the presence of enhanced microturbulence generated by energy release. These results imply that unresolved reconnection processes driven by small-scale flux emergence and cancellation may be responsible for such localized heating and small-scale jet activity. Furthermore, the temporal variation of the 304 Å flux observed in the SF region (in Figure 2(e)) indicates that the local footpoint heating could be a random and recurrent phenomenon, possibly modulated by turbulent photospheric motions. The advent of new solar instruments, including the Daniel K. Inouye Solar Telescope and the Chinese Giant Solar Telescope, will provide higher resolution imaging and spectroscopic observations, enabling us to confirm this possibility in the future.

5. Summary

Utilizing multiwavelength, high-resolution imaging and spectroscopic observations from NVST, SDO, and IRIS, we examined the nature and origin of counterstreaming flows within a forming AR filament. This work elucidates the

multithermal origin of counterstreaming flows within a forming filament and provides evidence of localized chromospheric heating at the filament footpoints. Our main findings are listed as follows:

1. The forming filament exhibits long-lived and multithermal counterstreaming flows, persisting for at least several hours in our observations. Our results reveal these flows are primarily interleaved unidirectional mass flows in opposing directions. In the H α passband, their typical speeds are around 20 km s⁻¹, while in the UV Si IV 1400 Å and EUV 304 Å passbands, their flow speeds can reach up to 40 km s⁻¹. IRIS spectral diagnostics show enhanced micro-turbulence in the filament counterstreaming flow regions, as indicated by the significant nonthermal width velocity of ~ 40 km s⁻¹ in Si IV 1400 Å spectral profiles.
2. Imaging evidence reveals these multithermal flows emanating from small-scale brightenings at the filament's ends. At their nascent phase, these flows manifest as a series of small-scale, collimated jets at the filament's ends. They continuously inject cool chromospheric and warm transition region plasma into the filament channel, thereby continuously supplying multitemperature plasma into the filament's interior and driving the counterstreaming flows. The averaged jet speeds detected at the filament's end are about 30 km s⁻¹ in H α and 44 km s⁻¹ in EUV/UV passbands, respectively.
3. At the small-scale brightenings at the filament's ends, where the multithermal jets originated, IRIS spectral diagnostics confirm significant chromospheric local heating. This is indicated by pronounced intensity enhancements and line broadening of the Si IV and C II spectral lines.
4. HMI observations reveal small-scale photospheric magnetic elements with opposing polarities undergo flux emergence and/or cancellation at the filament's ends, with a magnitude of 10^{17} Mx. These weak magnetic field activities suggest that unresolved magnetic reconnection events associated with flux cancellation may be driving these small-scale jets and associated localized footpoint heating.

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